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Cable tie

Abstract

A cable tie may include a desired length of belt, and a buckle connected to a proximal end portion of the belt. The buckle has a belt insertion hole and a slot continuous with the belt insertion hole. The belt insertion hole is configured such that the belt is longitudinally introduced thereinto. The slot is configured such that the belt is laterally introduced into the belt insertion hole therethrough. The slot includes an inlet opening that is gradually widened outward.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3339246	12/1966	Geisinger	24/16PB	B65D 63/1063
3339247	12/1966	Geisinger	24/16PB	B65D 63/1027
3837047	12/1973	Bunnell	24/16PB	B65D 63/1072
4447934	12/1983	Anscher	248/74.3	H02G 3/30
4991265	12/1990	Campbell	24/129R	B65D 63/1027
5832567	12/1997	Edwards	24/30.5 P	G09F 3/037
5839169	12/1997	Edwards	24/17AP	G09F 3/037
5901416	12/1998	Mears	24/17AP	B65D 63/1063
6594867	12/2002	Pettersson	24/30.5 P	B65D 63/1027
10954047	12/2020	Cogliandro	N/A	B65D 63/1081
11191254	12/2020	Davis	N/A	A01K 27/001
11338973	12/2021	Yamane	N/A	B65D 63/1063
2005/0204515	12/2004	Hewes	24/16PB	B65D 63/1063
2008/0028576	12/2007	Okamoto	24/16R	B65D 63/1027
2018/0080580	12/2017	Coulter	N/A	F16L 3/2336
2019/0009958	12/2018	Wiles	N/A	B65D 63/1072

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2001-208017	12/2000	JP	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to Japanese Patent Application serial number 2022-120460 filed Jul. 28, 2022, the contents of which are incorporated herein by reference.

BACKGROUND

(2) The present disclosure relates to a cable tie (a clamping band). More particularly, the present disclosure relates to a cable tie for tying together a bundle of tied articles (e.g., a wiring harness

that is used for vehicle wiring).

(3) Generally, a cable tie for tying together a bundle of tied articles (e.g., a wiring harness) includes a desired length of flexible belt that is configured to be wound or wrapped around the bundle of tied articles, and a buckle connected to a proximal end of the belt. In order to tie together the bundle of tied articles, the belt is wound or wrapped around the bundle of tied articles and is then inserted into a belt insertion hole formed in the buckle. Thereafter, the belt is fastened and is then locked by engaging an engagement strip formed in the belt insertion hole of the buckle. Thus, the bundle of tied articles is tied together by the cable tie.

(4) A known cable tie is taught by, for example, Japanese Laid-Open Patent Publication No. 2001-208017. The known cable tie includes an open-ended box-shaped buckle and a belt. The buckle includes a belt insertion hole longitudinally formed therethrough and having an inlet end. The belt insertion hole is configured such that the belt is longitudinally introduced thereinto through the inlet end. Further, the buckle includes a slot longitudinally formed therein. The slot extends along the entire length of the buckle so as to be continuous with the belt insertion hole over the entire length thereof. The longitudinal slot thus formed allows the belt to be laterally introduced into the belt insertion hole as necessary. Therefore, the belt can be introduced into the belt insertion hole not only through the inlet end of the belt insertion hole but also through the slot formed in the buckle.

(5) According to the known cable tie thus constructed, the belt can be effectively introduced into the belt insertion hole because the belt can be laterally introduced into the belt insertion hole through the longitudinal slot formed in the buckle. However, the known cable tie has room for improvement in structure or shape of the buckle.

(6) Thus, there is a need in the art for improved cable ties.

SUMMARY

(7) For example, in one aspect of the present disclosure, a cable tie may include a desired length of belt, and a buckle connected to a proximal end portion of the belt. The buckle has a belt insertion hole and a slot continuous with the belt insertion hole. The belt insertion hole is configured such that the belt is longitudinally introduced thereinto. The slot is configured such that the belt is laterally introduced into the belt insertion hole therethrough. The slot includes an inlet opening that is gradually widened outward.

(8) According to this aspect, the slot formed in the buckle includes the inlet opening that is gradually widened outward. Therefore, the belt can be easily introduced into the slot with the aid of such an inlet opening. As a result, the belt can be easily and smoothly introduced into the belt insertion hole of the buckle through the slot.

(9) Additional objects, features and advantages of the present disclosure will be readily understood after reading the following detailed description together with the accompanying drawings and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is an overall perspective view of a cable tie according to a first representative embodiment of the present disclosure, which is viewed from above;

(2) FIG. 2 is an overall perspective view of the cable tie, which is viewed from the bottom;

(3) FIG. 3 is an enlarged perspective view of a buckle of the cable tie, which is viewed from above;

(4) FIG. 4 is a side view of the cable tie, which is viewed from the right;

(5) FIG. 5 is a plan view of FIG. 3;

(6) FIG. 6 is a cross-sectional view taken along line VI-VI in FIG. 5;

(7) FIG. 7 is a side view of the cable tie, which illustrates a process of tying together a bundle of tied articles using the cable tie;

- (8) FIG. **8** is a side view of the cable tie, which illustrates the process of tying together the bundle of tied articles using the cable tie;
- (9) FIG. **9** is an enlarged partially cross-sectional plan view of the buckle into which a belt of the cable tie is introduced, which illustrates the process of tying together the bundle of tied articles using the cable tie;
- (10) FIG. **10** is a side view of a cable tie according to a second representative embodiment of the present disclosure, which is viewed from the right;
- (11) FIG. **11** is a side view of the cable tie, which illustrates a process of tying together a bundle of tied articles using the cable tie;
- (12) FIG. **12** is a side view of the cable tie, which illustrates the process of tying together the bundle of tied articles using the cable tie;
- (13) FIG. **13** is a side view of the cable tie, which illustrates the process of tying together the bundle of tied articles using the cable tie;
- (14) FIG. **14** is a side view of a cable tie according to a third representative embodiment of the present disclosure, which is viewed from the right;
- (15) FIG. **15** is a side view of a cable tie according to a fourth representative embodiment of the present disclosure, which is viewed from the right;
- (16) FIG. **16** is a view similar to FIG. **9**, which illustrates a belt of a cable tie according to a first modified embodiment of the first embodiment;
- (17) FIG. **17** is an overall perspective view of a cable tie according to a second modified embodiment of the first embodiment, which is viewed from above;
- (18) FIG. **18** is a schematic partial perspective view of the belt of the cable tie according to the second modified embodiment, which shows a modified form of a removed portion formed in the belt;
- (19) FIG. **19** is a schematic partial perspective view of the belt of the cable tie according to the second modified embodiment, which shows a modified form of the removed portion formed in the belt;
- (20) FIG. **20** is a schematic partial perspective view of the belt of the cable tie according to the second modified embodiment, which shows a modified form of the removed portion formed in the belt;
- (21) FIG. **21** is a view similar to FIG. **3**, which illustrates a buckle of a cable tie according to a third modified embodiment of the first embodiment,
- (22) FIG. **22** is a schematic partial perspective view of the buckle of the cable tie according to the third modified embodiment, which schematically shows a form of an inclined surface formed on the buckle;
- (23) FIG. **23** is a schematic partial perspective view of the buckle of the cable tie according to the third modified embodiment, which schematically shows a modified form of the inclined surface formed on the buckle;
- (24) FIG. **24** is a schematic partial perspective view of the buckle of the cable tie according to the third modified embodiment, which schematically shows a modified form of the inclined surface formed on the buckle;
- (25) FIG. **25** is a schematic partial perspective view of the buckle of the cable tie according to the third modified embodiment, which schematically shows a modified form of the inclined surface formed on the buckle.

DETAILED DESCRIPTION

- (26) Detailed representative embodiments of the present disclosure are shown in FIGS. **1** to **25**.
- (27) A first detailed representative embodiment of the present disclosure will be described with reference to FIGS. **1** to **9**. The first embodiment may be directed to a cable tie **1** for tying together a bundle of tied articles. Further, in this embodiment, a wiring harness **10** that is used for vehicle wiring may be shown as an example of the bundle of tied article. Further, a longitudinal or front-

back direction (forward and backward directions), a lateral direction (rightward and leftward directions) and a vertical direction (upward and downward directions) described therein are determined only for the purpose of convenience and not intended to limit the disclosure.

(28) As shown in FIGS. **1**, **2** and **4**, the cable tie **1** may be integrally formed as a unit by integral molding of a rigid synthetic resin. The cable tie **1** may include a desired length of flexible belt **2** and a buckle **4**. The belt **2** may have a substantially constant width over the entire length thereof and may be configured to be wrapped around the wiring harness **10**. The belt **2** may have a base portion **20** and a functional portion (not labeled). The base portion **20** of the belt **2** may be connected to the buckle **4** (which will be hereinafter described) at a proximal end **21** thereof. The base portion **20** may substantially horizontally linearly extend forward from the buckle **4**. Further, the proximal end **21** of the base portion **20** may be referred to as a proximal end portion of the belt **2**.

(29) As shown in FIGS. **1**, **2** and **4**, the functional portion of the belt **2** may include a first linear portion **25**, a second linear portion **27** and a third linear portion **29**. The first linear portion **25** may be connected to a distal end **22** of the base portion **20** at a proximal end (not labeled) thereof. Further, as shown in FIG. **4**, the first linear portion **25** may be inclined at an (first) angle B with respect to the base portion **20** in a direction which the belt **2** is wrapped around the wiring harness **10**. Further, such a direction may be hereinafter referred to as a belt wrapping direction W. In other words, the first linear portion **25** may be inclined with respect to the base portion **20** in the belt wrapping direction W so as to form an angle A (which may be referred to as a supplementary angle of the angle B) therebetween. Thus, the first linear portion **25** may be inwardly obliquely connected to the base portion **20** via a bent portion **37**.

(30) As shown in FIGS. **1**, **2** and **4**, the second linear portion **27** may be connected to a distal end **26** of the first linear portion **25** at a proximal end (not labeled) thereof. Further, as shown in FIG. **4**, the second linear portion **27** may be inclined at an (second) angle D with respect to the first linear portion **25** in the belt wrapping direction W. In other words, the second linear portion **27** may be inclined with respect to the first linear portion **25** in the belt wrapping direction W so as to form an angle C (which may be referred to as a supplementary angle of the angle D) therebetween. Thus, the second linear portion **27** may be inwardly obliquely connected to the first linear portion **25** via a bent portion **38**.

(31) As shown in FIGS. **1**, **2** and **4**, the third linear portion **29** may be connected to a distal end **28** of the second linear portion **27** at a proximal end (not labeled) thereof. Further, as shown in FIG. **4**, the third linear portion **29** may be inclined at an (third) angle F with respect to the second linear portion **27** in the belt wrapping direction W. In other words, the third linear portion **29** may be inclined with respect to the second linear portion **27** in the belt wrapping direction W so as to form an angle E (which may be referred to as a supplementary angle of the angle F) therebetween. Thus, the third linear portion **29** may be inwardly obliquely connected to the second linear portion **27** via a bent portion **39**.

(32) The first angle B, the second angle D and the third angle F may be determined to meet the following requirements: 1) the first angle B ≤ the second angle D ≤ the third angle F 2) the first angle B + the second angle D + the third angle F ≤ 90 degrees.

Further, in this embodiment, the first angle B, the second angle D and the third angle F are respectively determined to 10 degrees, 20 degrees and 30 degrees.

(33) As shown in FIGS. **1** and **3**, the base portion **20** of the belt **2** may have a plurality of engagement teeth or rack teeth **24** that are formed in an inner surface **23** thereof. The rack teeth **24** may preferably be arranged at a constant pitch distance in a longitudinal direction of the base portion **20**. Further, the first linear portion **25** of the functional portion of the belt **2** may be specially shaped in cross section. That is, a right (first) side edge **33** of the first linear portion **25** may have a special shape different from a left (second) side edge **35** of the first linear portion **25**. In particular, the right side edge **33** of the first linear portion **25** may have a lower rounded surface

33a and an upper slanted surface **33b** (FIG. 9). Thus, the right side edge **33** of the first linear portion **25** may be formed into a wedged or tapered shape as a whole. Conversely, the left side edge **35** of the first linear portion **25** may be simply chamfered so as to have a sheer or perpendicular end surface **36**. Further, the second linear portion **27** and the third linear portion **29** of the functional portion of the belt **2** may preferably be formed such that their right side edges (not labeled) respectively have a wedged or tapered shape. Similarly, the base portion **20** of the belt **2** may be formed such that its right side edges (not labeled) has a wedged or tapered shape as necessary. Further, the right side edge **33** and the left side edge **35** of the first linear portion **25** may be referred to as the right side edges **33** and the left side edge **35** of the belt **2** in some situation.

(34) As shown in FIG. 1, the third linear portion **29** of the belt **2** may have a plurality of gear tooth-shaped projections **32** that are formed in a central portion of an inner surface **31** thereof. The projections **32** may preferably be arranged at a constant pitch distance in a longitudinal direction of the third linear portion **29**.

(35) As shown in FIG. 4, in this embodiment, a length **L2** of the first linear portion **25**, a length **L3** of the second linear portion **27** and a length **L4** of the third linear portion **29** may be determined to be equal to each other. Further, a length of the functional portion of the belt **2** (i.e., a sum of the length **L2** of the first linear portion **25**, the length **L3** of the second linear portion **27** and the length **L4** of the third linear portion **29**, or a difference between an overall length **L** of the belt **2** and a length **L1** of the base portion **20** of the belt **2**) may be determined to be equal to 28 ± 10 percent of the overall length **L** of the belt **2**.

(36) Next, the buckle **4** will be described. As shown in FIGS. 3 and 5, the buckle **4** may have an open-ended box-shaped member having a belt insertion hole **53** that is centrally vertically (longitudinally) formed therethrough. The belt insertion hole **53** may be configured such that the belt **2** is introduced thereinto. As shown in FIG. 6, the belt insertion hole **53** may have an inlet end **54** and an outlet end **55** respectively opening to the outside. The belt insertion hole **53** may be configured such that the belt **2** (a distal end **30** of the third linear portion **29**) is inserted thereinto through the inlet end **54**.

(37) As shown in FIGS. 3 and 6, the buckle **4** may include a front wall **40**, a rear wall **42**, a right (first) side wall **44** and a left (second) side wall **49**. As shown in FIG. 6, the front wall **40** may be connected to the proximal end **21** of the base portion **20** of the belt **2** via an upwardly curved concave quadrant arch-shaped thinned portion **40a** formed on an upper end thereof. Further, the buckle **4** may include an engagement strip **60** formed on a vertical inner surface **41** of the front wall **40**. The engagement strip **60** may be configured to flex back and forth with respect to the front wall **40**. The engagement strip **60** may have an engagement claw **61** formed thereon. The engagement claw **61** may be configured to engage the rack teeth **24** formed on the base portion **20** of the belt **2** when the belt **2** is inserted into the belt insertion hole **53** of the buckle **4** in order to tie together the wiring harness **10**. Further, as shown in FIG. 5, the engagement claw **61** may have a rounded portion **61a** formed in a left edge thereof.

(38) As shown in FIGS. 5 and 6, the rear wall **42** may be positioned opposite to the front wall across the belt insertion hole **53**. The rear wall **42** may have a laterally rounded surface **43** formed in an upper end thereof and facing the belt insertion hole **53**. The rounded surface **43** may preferably have the substantially same width as a width of the belt **2**. The rounded surface **43** may contribute to easy insertion of the belt **2** into the belt insertion hole **53**.

(39) As shown in FIG. 5, the right side wall **44** may be connected to a right end of the front wall **40** and a right end of the rear wall **42**. As shown in FIG. 1, the right side wall **44** may include a laterally outwardly projected horizontal (first) projected portion **47** that is formed on an outer surface thereof so as to extend along an upper end periphery thereof. Further, the right side wall **44** may include a laterally outwardly projected vertical (second) projected portion **48** formed on the outer surface thereof and integrated with the horizontal projected portion **47**. Thus, as shown in FIG. 5, the right side wall **44** may have a thickness **T1** that is substantially greater than a thickness

T2 of the rear end wall 42 and a thickness T3 of the left side wall 49.

(40) As shown in FIG. 3, the left side wall 49 may be positioned laterally opposite to the right side wall 44. The left side wall 49 may be connected to a left end of the front wall 40 and a left end of the rear wall 42. As shown in FIGS. 3 and 5, the left side wall 49 may be vertically divided in a central portion thereof, so as to be formed into two (front and rear) separate wall portions (not labeled). Thus, the left side wall 49 may have a vertical slot 56 formed between the front wall portion and the rear wall portion. The vertical slot 56 may extend along the entire length of the left side wall 49 so as to be continuous with the belt insertion hole 53 over the entire length thereof. The vertical slot 56 thus formed allows the belt 2 to be laterally introduced into the belt insertion hole 53 as necessary. Therefore, the belt 2 may be introduced into the belt insertion hole 53 not only through the inlet end 54 of the belt insertion hole 53 but also through the vertical slot 56 formed in the left side wall 49.

(41) As shown in FIG. 5, the rear wall portion of the left side wall 49 may be specially shaped. In particular, the rear wall portion may have a projection or overhang 50 formed therealong and projecting into the belt insertion hole 53. The overhang 50 may have an inclined (inner) surface 51 hanging over and facing the belt insertion hole 53. Further, the overhang 50 may preferably have a height H equal to approximately 80 percent of a height (thickness) H1 of the left side edge of the belt 2.

(42) As shown in FIG. 5, the front and rear wall portions of the left side wall 49 may respectively have a front vertically elongated projection 52a and a rear vertically elongated projection 52b. The front projection 52a and the rear projection 52b may respectively have vertical surfaces 59a and 59b facing each other. Further, the vertical surface 59a of the front vertical projection 52a may be inclined or curved frontward and leftward (outward) while the vertical surface 59b of the rear vertical projection 52b may be inclined or curved rearward and leftward (outward). Thus, the vertical slot 56 formed in the left side wall 49 may be gradually widened leftward (outward), so as to have a flared inlet opening 57.

(43) Next, a standard operation for tying together the wiring harness 10 using the cable tie 1 will be described. First, the belt 2 may be applied to the wiring harness 10 from below so as to intersect with the wiring harness 10 at a substantially right angle. The belt 2 may then be looped around the wiring harness 10 in the belt wrapping direction W. Thereafter, the distal end 30 of the third linear portion 29 of the belt 2 looped around the wiring harness 10 may be introduced into the belt insertion hole 53 of the buckle 4 through the inlet end 54 of the belt insertion hole 53. Next, the distal end 30 of the third linear portion 29 introduced into the belt insertion hole 53 may be pulled and drawn from the belt insertion hole 53 through the outlet end 55 of the belt insertion hole 53. As a result, the third linear portion 29, the second linear portion 27 and the first linear portion 25 of the belt 2 may be sequentially introduced into the belt insertion hole 53. When the distal end 30 of the third linear portion 29 of the belt 2 is further pulled, the base portion 20 of the belt 2 may be introduced into the belt insertion hole 53 while flexing the engagement strip 60 formed in the belt insertion hole 53 back and forth. As a result, the belt 2 (the base portion 20) may be gradually fastened while being wrapped around the wiring harness 10. Thereafter, when the pulling operation of the belt 2 is stopped after the belt 2 is fastened to some extent, the engagement claw 61 formed on the engagement strip 60 may elastically engage one of the rack teeth 24 formed on the base portion 20 of the belt 2, so that the belt 2 may be fastened and locked by the buckle 4. Thus, the wiring harness 10 may be temporarily or substantially tied together by the cable tie 1. Further, as previously described, the buckle 4 is connected to the belt 2 via the thinned portion 40a. Therefore, when the belt 2 is fastened, stress produced in the belt 2 may be effectively reduced or dispersed by the thinned portion 40a.

(44) Preferably, the belt 2 may be further fastened by using a specialized fastening tool (not shown) such that the wiring harness 10 can be completely tied together by the cable tie 1. In particular, a mouthpiece or tool head of the specialized tool may be fitted to the outlet end 55 of the belt

insertion hole **53**. Thereafter, the fastening tool may be operated, so as to further fasten the belt **2**. When the belt **2** is sufficiently fastened, the engagement claw **61** of the buckle **4** may elastically engage another of the rack teeth **24** of the belt **2**, so that the belt **2** may be sufficiently fastened and ultimately locked by the buckle **4**. Thus, the wiring harness **10** may be securely tied together by the cable tie **1**. Further, the belt **2** may be ultimately fastened by hand without using the fastening tool as necessary.

(45) Generally, after the belt **2** is locked by the buckle **4**, an excess portion of the belt **2** (i.e., a distal portion (not labeled) of the base portion **20** including the functional portion) drawn from the outlet end **55** of the belt insertion hole **53** of the buckle **4** may preferably be cut off by using cutting tools, e.g., nippers or scissors (not shown).

(46) The wiring harness **10** is sometimes positioned adjacent to a wall-like obstacle **70** (e.g., a structural member such as a vehicle body panel). Next, an operation for tying together the wiring harness **10** using the cable tie **1** in such a case will be hereinafter described with reference to FIGS. **4** and **7** to **9**. As shown in FIG. **4**, first, the belt **2** may be applied to the wiring harness **10** from below in such a manner that the belt **2** intersects with the wiring harness **10** at a substantially right angle. In particular, the belt **2** may be positioned under the wiring harness **10** in such a manner that the distal end **30** of the third linear portion **29** of the belt **2** may contact the obstacle **70**. At this time, the third linear portion **29** may be inclined at an angle with respect to the obstacle **70**.

(47) When the belt **2** is longitudinally pressed to the obstacle **70** in this condition (the condition shown in FIG. **4**), a pressing force applied to the belt **2** may substantially be concentrated to the bent portion **39** formed between the third linear portion **29** and the second linear portion **27**. As a result, the third linear portion **29** may come into contact with the obstacle **70** while relatively flexing with respect to the second linear portion **27** (solid lines in FIG. **7**). Upon contact of the third linear portion **29** with the obstacle **70**, the distal end **28** of the second linear portion **27** may come into contact with the obstacle **70** in a condition in which while the second linear portion **27** is inclined at an angle with respect to the obstacle **70**.

(48) When the belt **2** is further pressed to the obstacle **70** in this condition, the pressing force applied to the belt **2** may substantially be concentrated to the bent portion **38** formed between the second linear portion **27** and the first linear portion **25**. As a result, the second linear portion **27** may come into contact with the obstacle **70** while relatively flexing with respect to the first linear portion **25** (shown by chain double-dashed lines in FIG. **7**). At this time, as shown by chain double-dashed lines in FIG. **7**, due to the pressing force applied to the belt **2**, the base portion **20** of the belt **2** may be flexed or deflected upward, i.e., in the belt wrapping direction **W**, in a middle portion thereof. Therefore, the second linear portion **27** may come into contact with the obstacle **70** while moving or sliding upward along the obstacle **70**. Upon contact of the second linear portion **27** with the obstacle **70**, the distal end **26** of the first linear portion **25** may come into contact with the obstacle **70** in a condition in which the first linear portion **25** is inclined at an angle with respect to the obstacle **70**. Conversely, the third linear portion **29** may be spaced from the obstacle **70** so as to obliquely extend from the obstacle **70** in the belt wrapping direction **W** because the third linear portion **29** is inclined at the angle **F** with respect to the second linear portion **27** in the belt wrapping direction **W**.

(49) When the belt **2** is further pressed to the obstacle **70** in this condition, the pressing force applied to the belt **2** may substantially be concentrated to the bent portion **37** formed between the first linear portion **25** and the base portion **20**. As a result, the first linear portion **25** may come into contact with the obstacle **70** while relatively flexing with respect to the base portion **20** (shown by solid lines in FIG. **8**). At this time, as shown by solid lines in FIG. **8**, due to the pressing force applied to the belt **2**, the base portion **20** may be arcuately flexed upward in the middle portion thereof. Therefore, the first linear portion **25** may come into contact with the obstacle **70** while moving or sliding upward along the obstacle **70**. Upon contact of the first linear portion **25** with the obstacle **70**, the distal end **22** of the base portion **20** may come into contact with the obstacle **70** in a

condition in which the base portion **20** is inclined at an angle with respect to the obstacle **70**. Conversely, the second linear portion **27** may be spaced from the obstacle **70** so as to obliquely extend from the obstacle **70** in the belt wrapping direction **W** because the second linear portion **27** is inclined at the angle **D** with respect to the first linear portion **25** in the belt wrapping direction **W**. (50) When the belt **2** is further pressed to the obstacle **70** in this condition, due to the pressing force applied to the belt **2**, the distal portion of the base portion **20** may come into contact with the obstacle **70** (shown by chain double-dashed lines in FIG. **8**). At this time, as shown by chain double-dashed lines in FIG. **8**, due to the pressing force applied to the belt **2**, the middle portion of the base portion **20** may be further arcuately flexed upward. Therefore, the distal portion of the base portion **20** may come into contact with the obstacle **70** while moving or sliding upward along the obstacle **70**. Upon contact of the distal portion of the base portion **20** with the obstacle **70**, the first linear portion **25** may be spaced from the obstacle **70** so as to obliquely extend from the obstacle **70** in the belt wrapping direction **W** because the first linear portion **25** is inclined at the angle **B** with respect to the base portion **20** in the belt wrapping direction **W**.

(51) Thus, when the belt **2** is longitudinally pressed to the obstacle **70** in the condition in which the distal end **30** of the third linear portion **29** of the belt **2** contacts the obstacle **70** (FIG. **4**), the third linear portion **29**, the second linear portion **27** and the first linear portion **25** may be pressed against and spaced away from the obstacle **70** in a stepwise fashion while sliding upward along the obstacle **70**, so as to continuously extend backward (i.e., toward the buckle **4**) from the obstacle **70** in this order. As shown by chain double-dashed lines in FIG. **8**, when the belt **2** is pressed to the obstacle **70** until the distal portion of the base portion **20** of the belt **2** comes into contact with the obstacle **70**, all of the third linear portion **29**, the second linear portion **27** and the first linear portion **25** may be spaced from the obstacle **70** so as to substantially arcuately extend backward (i.e., toward the buckle **4**) from the obstacle **70**. This means that the third linear portion **29**, the second linear portion **27** and the first linear portion **25** may extend in the belt wrapping direction **W** around the wiring harness **10** in this order. Further, each of the first linear portion **25**, the second linear portion **27** and the third linear portion **29** may have a substantial length. Therefore, the third linear portion **29**, the second linear portion **27** and the first linear portion **25** may extend backward from the obstacle **70** over a desired distance.

(52) According to the cable tie **1**, even if the wiring harness **10** is positioned adjacent to the obstacle **70**, a worker can easily pinch or hold the distal end **30** (which may be referred to as a distal end portion of the belt **2**) of the third linear portion **29** of the belt **2** applied to the wiring harness **10** from below by the fingers in order to wrap the belt **2** around the wiring harness **10**. Therefore, the worker can quickly wrap the belt **2** around the wiring harness **10** by holding the distal end **30** of the third linear portion **29**, thereby tying together the wiring harness **10**.

(53) Further, the distal end **30** of the third linear portion **29** of the belt **2** looped around the wiring harness **10** sometimes cannot be vertically introduced into the belt insertion hole **53** of the buckle **4** through the inlet end **54** of the belt insertion hole **53**. In such a case, in order to tie together the wiring harness **10** using the cable tie **1**, the belt **2** may be laterally introduced into the belt insertion hole **53** of the buckle **4** through the vertical slot **56** formed in the left side wall **49**. In particular, the belt **2** may be applied to the wiring harness **10** from below. The belt **2** may then be looped around the wiring harness **10** in the belt wrapping direction **W**. Thereafter, the first linear portion **25** of the belt **2** looped around the wiring harness **10** may be laterally introduced into the belt insertion hole **53** of the buckle **4** through the vertical slot **56** formed in the left side wall **49** (FIG. **9**). Thereafter, the wiring harness **10** may be tied together by the cable tie **1** in the same manner as the standard operation described above. Thus, even in the case where the distal end **30** of the third linear portion **29** of the belt **2** looped around the wiring harness **10** cannot be vertically introduced into the belt insertion hole **53** of the buckle **4** through the inlet end **54** of the belt insertion hole **53**, the wiring harness **10** can be easily and reliably tied together by the cable tie **1**.

(54) As previously described, the right side edge **33** of the first linear portion **25** may have the

lower rounded surface **33a** and the upper slanted surface **33b** (FIG. 9), so as to have the tapered shape as a whole. Therefore, the first linear portion **25** of the belt **2** may be easily and smoothly inserted into the vertical slot **56**. Further, the vertical slot **56** may have the flared inlet opening **57**. This may also contribute to easy insertion of the first linear portion **25** of the belt **2** into the vertical slot **56**. Thus, the first linear portion **25** of the belt **2** looped around the wiring harness **10** may be easily and smoothly introduced into the belt insertion hole **53** of the buckle **4** through the vertical slot **56** formed in the left side wall **49** of the buckle **4**.

(55) Further, as previously described, the overhang **50** formed in the rear wall portion **52b** of the left side wall **49** of the buckle **4** may have the height **H** equal to approximately 80 percent of the height **H1** of the right side edge **35** of the first linear portion **25** of the belt **2**. Therefore, the first linear portion **25** of the belt **2** introduced into the belt insertion hole **53** of the buckle **4** may be effectively prevented from being disengaged or slipped from the buckle **4** via the vertical slot **56**.

(56) According to the cable tie **1** of the first embodiment, the belt **2** may include the base portion **20** and the functional portion. The functional portion may include the first linear portion **25**, the second linear portion **27** and the third linear portion **29** that are connected to each other at the predetermined angles (i.e., the angle **D** and the angle **F**). Further, the first linear portion **25** of the functional portion may be connected to the base portion **20** at the predetermined angle (i.e., the angle **B**). Therefore, in the case where the wiring harness **10** is positioned adjacent to the obstacle **70**, when the belt **2** positioned under the wiring harness **10** is pressed to the obstacle **70** after the distal end **30** of the third linear portion **29** of the belt **2** contacts the obstacle **70**, the third linear portion **29**, the second linear portion **27** and the first linear portion **25** may turn stepwise at the obstacle **70** in this order so as to gradually overhang the wiring harness **10** in the belt wrapping direction **W**. Further, the third linear portion **29** may be inclined at an angle equal to a sum of the angles **B**, **D** and **F** with respect to the obstacle **70**. Therefore, even if the wiring harness **10** is positioned adjacent to the obstacle **70**, the worker can easily and quickly wrap the belt **2** around the wiring harness **10** by holding the distal end **30** of the third linear portion **29**.

(57) Further, the first angle **B**, the second angle **D** and the third angle **F** are respectively determined to 10 degrees, 20 degrees and 30 degrees. In addition, the length **L2** of the first linear portion **25**, the length **L3** of the second linear portion **27** and the length **L4** of the third linear portion **29** are determined to be substantially equal to each other. Therefore, the third linear portion **29**, the second linear portion **27** and the first linear portion **25** (i.e., the functional portion of the belt **2**) may be well-balanced in structure and function.

(58) Further, the sum of the length **L2** of the first linear portion **25**, the length **L3** of the second linear portion **27** and the length **L4** of the third linear portion **29** (i.e., the length of the functional portion of the belt **2**) may be determined to be equal to 28 ± 10 percent of the overall length **L** of the belt **2**. Such a ratio of the length of the functional portion of the belt **2** to the overall length **L** of the belt **2** may contribute to an increase in function of the functional portion of the belt **2** and an increase in ease of handling of the belt **2**.

(59) Further, the buckle **4** may include the vertical slot **56** that is continuous with the belt insertion hole **53**. Therefore, the belt **2** can be laterally introduced into the belt insertion hole **53** of the buckle **4** through the vertical slot **56** instead of the inlet end **54**. Further, the vertical slot **56** may have the flared inlet opening **57**. Therefore, the belt **2** can be easily and smoothly introduced into the belt insertion hole **53** of the buckle **4** through the vertical slot **56**.

(60) Further, the right side edge **33** of the belt **2** may have the tapered shape. Therefore, the belt **2** can be easily and smoothly introduced into the belt insertion hole **53** of the buckle **4** through the vertical slot **56**.

(61) Further, the buckle **4** may include the right side wall **44** having the thickness **T1** greater than the thickness **T2** of the rear end wall **42** and the thickness **T3** of the left side wall **49**. Therefore, the buckle **4** may have a desired strength regardless of the presence of the vertical slot **56** formed in the left side wall **49**.

(62) Further, the left side wall **49** of the buckle **4** may have the overhang **50** formed in rear wall portion and hanging over the belt insertion hole **53**. The overhang **50** may have the inclined surface **51** facing the belt insertion hole **53**. Further, the overhang **50** may have the height H equal to approximately 80 percent of the height H1 of the left side edge **35** of the belt **2**. Therefore, the overhang **50** may effectively interfere with the left side edge **35** of the belt **2** introduced into the belt insertion hole **53**, thereby preventing the belt **2** from being disengaged or slipped from the buckle **4** through the vertical slot **56**.

(63) Further, the vertical slot **56** of the buckle **4** may have the flared inlet opening **57**. Therefore, the belt **2** can be easily introduced into the belt insertion hole **53** of the buckle **4** through the vertical slot **56**.

(64) Next, a second detailed representative embodiment will now be described with reference to FIGS. **10** to **13**. Further, because the second embodiment relates to the first embodiment, only the constructions and elements that are different from the first embodiment will be explained in detail. Elements that are the same in the first and second embodiments will be identified by the same reference numerals and a detailed description of such elements may be omitted.

(65) Similar to the cable tie **1** of the first embodiment, a cable tie **101** of the second embodiment may be integrally formed as a unit by integral molding of a rigid synthetic resin. The cable tie **101** may include a desired length of flexible belt **102** and the buckle **4**. The cable tie **101** may be different from the cable tie **1** of the first embodiment in that the belt **102** is structurally different from the belt **2** of the first embodiment. In particular, as shown in FIGS. **10** to **13**, unlike the first embodiment, the functional portion of the belt **102** may include only the first linear portion **25** and the second linear portion **27**. Further, in this embodiment, the second linear portion **27** may have the gear tooth-shaped projections **32** that are formed in a central portion of an inner surface thereof. The projections **32** may preferably be arranged at a constant pitch distance in a longitudinal direction of the second linear portion **27**.

(66) As shown in FIG. **10**, similar to the first embodiment, the first linear portion **25** may be inclined at the (first) angle B with respect to the base portion **20** in the belt wrapping direction W. Further, the second linear portion **27** may be inclined at the (second) angle D with respect to the first linear portion **25** in the belt wrapping direction W.

(67) However, in this embodiment, the first angle B and the second angle D may respectively be determined to 30 degrees. Generally, the angle B and the angle D may be determined to meet the following requirements: 1) the angle B ≤ the angle D 2) the angle B + the angle D ≤ 90 degrees.

(68) As shown in FIG. **10**, in this embodiment, the length L2 of the first linear portion **25** is determined to be greater than the length L3 of the second linear portion **27**. Further, the length of the functional portion of the belt **102** (i.e., a sum of the length L2 of the first linear portion **25** and the length L3 of the second linear portion **27**, or a difference between the overall length L of the belt **102** and the length L1 of the base portion **20** of the belt **102**) may be determined to be equal to 28±10 percent of the overall length L of the belt **102**. Further, the length L2 of the first linear portion **25** may be determined to be equal to or smaller than the length L3 of the second linear portion **27** as necessary.

(69) According to the cable tie **101**, in a condition where the belt **102** is positioned under the wiring harness **10** with the distal end **28** of the second linear portion **27** contacting the obstacle **70** (FIG. **10**), when the belt **102** is longitudinally pressed to the obstacle **70**, the second linear portion **27** and the first linear portion **25** may be pressed against and spaced away from the obstacle **70** in a stepwise fashion while sliding upward along the obstacle **70**, so as to continuously extend backward (i.e., toward the buckle **4**) from the obstacle **70** in this order (FIGS. **11** to **13**). As shown in FIG. **13**, when the belt **102** is pressed to the obstacle **70** until the distal portion of the base portion **20** of the belt **102** comes into contact with the obstacle **70**, both of the second linear portion **27** and the first linear portion **25** may be spaced away from the obstacle **70** so as to substantially arcuately extend backward (i.e., toward the buckle **4**) from the obstacle **70**. This means that the

second linear portion **27** and the first linear portion **25** may extend in the belt wrapping direction **W** around the wiring harness **10** in this order. Further, each of the first linear portion **25** and the second linear portion **27** may have a substantial length. Therefore, the second linear portion **27** and the first linear portion **25** may extend backward from the obstacle **70** over a desired distance.

(70) Further, according to the cable tie **101**, even if the wiring harness **10** is positioned adjacent to the obstacle **70**, the worker can easily pinch or hold the distal end **28** (which may be referred to as a distal end portion of the belt **102**) of the second linear portion **27** of the belt **102** applied to the wiring harness **10** from below by the fingers in order to wrap the belt **102** around the wiring harness **10**. Therefore, the worker can quickly wrap the belt **102** around the wiring harness **10** by holding the distal end **28** of the second linear portion **27**, thereby tying together the wiring harness **10**. Further, even in a case where the distal end **28** of the second linear portion **27** of the belt **102** looped around the wiring harness **10** cannot be vertically introduced into the belt insertion hole **53** of the buckle **4** through the inlet end **54** of the belt insertion hole **53**, the first linear portion **25** of the belt **102** may be laterally introduced into the belt insertion hole **53** of the buckle **4** through the vertical slot **56** formed in the left side wall **49**. Therefore, even in such a case, the wiring harness **10** can be easily and reliably tied together by the cable tie **101**.

(71) Next, a third detailed representative embodiment will now be described with reference to FIG. **14**. Further, because the third embodiment relates to the second embodiment, only the constructions and elements that are different from the first embodiment will be explained in detail. Elements that are the same in the first and second embodiments will be identified by the same reference numerals and a detailed description of such elements may be omitted.

(72) A cable tie **201** of the third embodiment may include a desired length of flexible belt **202** and the buckle **4**. The cable tie **201** may be different from the cable tie **101** of the second embodiment in that the belt **202** is structurally different from the belt **102** of the second embodiment. In particular, unlike the second embodiment, the first angle **B** may be determined to 25 degrees. The cable tie **201** may have the substantially same function as the cable tie **101** of the second embodiment.

(73) Next, a fourth detailed representative embodiment will now be described with reference to FIG. **15**. Further, because the fourth embodiment relates to the first embodiment, only the constructions and elements that are different from the first embodiment will be explained in detail. Elements that are the same in the first and fourth embodiments will be identified by the same reference numerals and a detailed description of such elements may be omitted.

(74) A cable tie **301** of the fourth embodiment may include a desired length of flexible belt **302** and the buckle **4**. The cable tie **301** may be different from the cable tie **1** of the first embodiment in that the belt **302** is structurally different from the belt **2** of the first embodiment. In particular, unlike the first embodiment, each of the second angle **D** and the third angle **F** may be determined to 10 degrees. The cable tie **301** may have the substantially same function as the cable tie **1** of the first embodiment.

(75) Naturally, various changes and modifications may be made to the present disclosure without departing from the scope of the disclosure. For example, in the first embodiment, the functional portion of the belt **2** may include a fourth linear portion connected to the distal end **30** of the third linear portion **29** and inclined at a fourth angle with respect to the third linear portion **29**. In such a case, the first angle **B**, the second angle **D**, the third angle **F** and the fourth angle may be determined to meet the requirements “the first angle $B \leq$ the second angle $D \leq$ the third angle $F \leq$ the fourth angle” and “the first angle $B +$ the second angle $D +$ the third angle $F +$ the fourth angle ≤ 90 degrees.” Further, the functional portion of the belt **2** may include additional linear portions in addition to the fourth linear portion.

(76) Further, in the first embodiment, the length **L2** of the first linear portion **25**, the length **L3** of the second linear portion **27** and the length **L4** of the third linear portion **29** may be determined to be equal to each other. However, the length **L2**, the length **L3** and the length **L4** may be determined

to be different from each other at a difference of up to 10 percent.

(77) Further, in the first embodiment, the left side edge **35** of the first linear portion **25** may have the perpendicular end surface **36** (FIG. **9**). However, as shown in FIG. **16** (a first modified embodiment), the left side edge **35** of the first linear portion **25** may have a slanted end surface **36'** that is configured to be parallel to the inclined surface **51** of the overhang **50** when the belt **2** is introduced into the belt insertion hole **53** of the buckle **4**. According to the modified embodiment of the first embodiment, the belt **2** introduced into the belt insertion hole **53** of the buckle **4** may be more effectively prevented from being disengaged or slipped from the buckle **4** through the vertical slot **56**. Similarly, in each of the second to fourth embodiment, the left side edge **35** of the first linear portion **25** may be modified in the same manner as the first embodiment.

(78) The first embodiment may be further modified. In particular, as shown in FIG. **17** (a second modified embodiment), the right side edge **33** of the belt **2** may be partially removed in the first linear portion **25**, thereby forming a cutout or removed portion **25a** that penetrates the first linear portion **25** in a belt thickness direction. The removed portion **25a** may serve as a sign that indicates a position of the first linear portion **25**. Therefore, the worker can easily and reliably introduce the first linear portion **25** into the belt insertion hole **53** of the buckle **4** through the vertical slot **56** with the help of the removed portion **25a** even when the worker cannot see the first linear portion **25**. Similarly, in each of the second to fourth embodiment, the right side edge **33** of the first linear portion **25** may be modified in the same manner as the first embodiment.

(79) Further, the removed portion **25a** formed in the first linear portion **25** may be variously changed in shape. For example, as shown in FIG. **18**, the removed portion **25a** may be changed to a shallow rectangular recessed portion **25a'** that does not penetrate the first linear portion **25** in the belt thickness direction. Further, as shown in FIG. **19**, the removed portion **25a** may be changed to a semicircular recessed portion **25a''** that does not penetrate the first linear portion **25** in the belt thickness direction. Further, as shown in FIG. **20**, the removed portion **25a** may be changed to a deep rectangular recessed portion **25a'''** that does not penetrate the first linear portion **25** in the belt thickness direction.

(80) The first embodiment may be further modified. In particular, as shown in FIGS. **21** and **22** (a third modified embodiment), at least one of the front wall portion (the front vertical projection **52a**) and the rear wall portion (the rear vertical projection **52b**) of the left side wall **49** of the buckle **4** may be modified. For example, the front wall portion of the left side wall **49** of the buckle **4** may have a triangular chamfered portion or inclined surface (D) that extends over an upper surface (A), an inner surface (B) and an outer surface (C) thereof. Further, the inner surface (B) may include the vertical surface **59a** of the front vertical projection **52a** formed in the front wall portion of the left side wall **49**. The inclined surface (D) may contribute to easy insertion of the first linear portion **25** of the belt **2** into the vertical slot **56**. Similarly, in each of the second to fourth embodiment, the buckle **4** may be modified in the same manner as the first embodiment.

(81) Further, the inclined surface (D) formed in the front wall portion of the left side wall **49** may be variously changed or modified in shape. For example, as shown in FIG. **23**, the inclined surface (D) may have a laterally elongated triangular shape. Further, as shown in FIG. **24**, the inclined surface (D) may have a rectangular shape. Further, as shown in FIG. **25**, the front wall portion of the left side wall **49** of the buckle **4** may have a cutout (E) instead of the inclined surface (D).

(82) Further, the cable tie **1** of the first embodiment may be modified. In particular, the buckle **4** may be provided with a connecting member (e.g., an anchor), so as to be connected to a vehicle body. According to the cable tie **1** thus modified, the wiring harness **10** tied together by the cable tie **1** can be secured to the vehicle body via the connecting member.

(83) Representative examples of the present disclosure have been described in detail with reference to the attached drawings. This detailed description is merely intended to teach a person of skill in the art further details for practicing preferred aspects of the present disclosure and is not intended to limit the scope of the disclosure. Only the claims define the scope of the claimed disclosure.

Therefore, combinations of features and steps disclosed in the foregoing detail description may not be necessary to practice the disclosure in the broadest sense, and are instead taught merely to particularly describe detailed representative examples of the disclosure. Moreover, the various features taught in this specification may be combined in ways that are not specifically enumerated in order to obtain additional useful embodiments of the present disclosure.

Claims

1. A cable tie, comprising: an elongated belt; and a buckle connected to a proximal end portion of the belt, wherein the buckle has a belt insertion hole extending longitudinally between an upper end of the buckle and a lower end of the buckle, and a slot continuous with the belt insertion hole, wherein the belt insertion hole is configured such that the belt is longitudinally introduced thereinto, wherein the slot is configured such that the belt is laterally introduced into the belt insertion hole therethrough, wherein the slot includes a flared inlet opening that is gradually widened outward, wherein the buckle includes a front wall connected to the belt, a rear wall positioned opposite to the front wall across the belt insertion hole, a first side wall connected to an end of the front wall and an end of the rear wall, and a second side wall positioned opposite to the first side wall, wherein an upper portion of each of the front wall, the rear wall, the first side wall and the second side wall define an upper opening of the belt insertion hole and a lower portion of each of the front wall, the rear wall, the first side wall and the second side wall define a lower opening of the belt insertion hole; and wherein the slot is formed in the second side wall.
 2. The cable tie as defined in claim 1, wherein the belt includes a first side edge and a second side edge positioned opposite to each other, and wherein the first side edge is formed into a tapered shape.
 3. The cable tie as defined in claim 1, wherein the first side wall has a thickness greater than a thickness of the rear end wall and a thickness of the second side wall.
 4. The cable tie as defined in claim 1, wherein the buckle includes a projection formed therealong and projecting into the belt insertion hole, and wherein the projection has an inner surface facing the belt insertion hole.
 5. The cable tie as defined in claim 4, wherein the belt includes a first side edge and a second side edge positioned opposite to each other, and wherein the second side edge has an end surface that is configured to be parallel to the inner surface of the projection when the belt is introduced into the belt insertion hole.
 6. The cable tie as defined in claim 5, wherein the projection has a height not less than 60 percent of a thickness of the second side edge of the belt.
 7. The cable tie as defined in claim 1, wherein the buckle has a pair of vertically elongated projections respectively having surfaces facing each other, and wherein the surfaces of the projections are respectively inclined outward, thereby defining the inlet opening of the slot.
 8. The cable tie as defined in claim 1, wherein the belt is configured to be wrapped around a bundle of tied articles, wherein the belt includes a base portion linearly extending from the buckle and a functional portion having a first linear portion and a second linear portion, wherein the first linear portion is connected to a distal end of the base portion at a proximal end thereof, wherein the second linear portion is connected to a distal end of the first linear portion at a proximal end thereof, wherein the first linear portion is inclined at a first angle with respect to the base portion in a belt wrapping direction, wherein the second linear portion is inclined at a second angle with respect to the first linear portion in the belt wrapping direction, and wherein the first angle is determined to be equal to or smaller than the second angle.
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